

- 9 (a) Define the mass defect of a nucleus.

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.....

..... [2]

- (b) Thorium-228 ($^{228}_{90}\text{Th}$) is radioactive.

Table 9.1 shows the mass defects of thorium-228 and some other nuclides.

Table 9.1

nuclide	mass defect/u
^4_2He	0.030 377
$^{224}_{88}\text{Ra}$	1.846 840
$^{228}_{89}\text{Ac}$	1.869 820
$^{228}_{90}\text{Th}$	1.871 290
$^{228}_{91}\text{Pa}$	1.866 129

- (i) Use the data in Table 9.1 to explain why thorium-228 is an α -emitter and **not** a β^- emitter or a β^+ emitter.

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..... [4]

- (ii) Calculate the energy, in MeV, released by the α -decay of one thorium-228 nucleus.
Give your answer to three significant figures.

energy = MeV [3]

[Total: 9]